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An interview with Kostas Kampourakis

By Juan Gefaell, PhD student at the Department of Biochemistry, Genetics, and Immunology, University of Vigo (gefaell@uvigo.es)



Kostas Kampourakis

Kostas Kampourakis is a science educator currently working at the University of Geneva. He is interested in the teaching and the public understanding of science, especially of evolution and genetics as well as of the nature of science. He has published dozens of scientific papers on various topics as well as authored and edited several books for specialized audiences and the general public, including *Turning Points: How Critical Events Have Driven Human Evolution, Life, and Development* (Prometheus Books, 2018), *Uncertainty: How It Makes Science Advance* (with Kevin McCain; Oxford University Press, 2019), *Understanding Evolution* (Cambridge University Press, 2020), and *Philosophy of Science for Biologists* (with Tobias Uller; Cambridge University Press, 2020). He is also editor of the Understanding Life book series by Cambridge University Press and has been the Editor-In-Chief of the journal *Science & Education* for several years. His latest book is titled *Ancestry Re-Imagined: Dismantling the Myth of Genetic Ethnicities*, forthcoming in Oxford University Press. As a result of his participation in EvoKE BCN 2021, Kostas has been kind enough to answer the following interview.

Q- You were formally trained as a biologist and science educator, but you have also made significant contributions to the philosophy of science and biology. What influences have shaped your work? What authors or approaches have had the most impact on you as an academic?

R- Indeed, I have no formal training in philosophy of science, yet I am the editor or co-editor of several books on the topic, and very interested in it. I think that philosophy of science is absolutely necessary, both for doing science and for teaching it. My research focuses on the latter, and I cannot think how one might teach about scientific concepts, explanations, models and theories (this is what we actually do in science education) without drawing on the philosophy of science.

Ever since I was an undergraduate student in biology, I had the feeling that something was missing from my studies. There was something that I needed in order to complete the picture and make sense of what I was studying. Luckily, a friend of mine who is older and was at the time very well versed in the philosophy of science guided me in my first foray in philosophy of science. Since then, philosophy of science has always been around anything I am doing.

Later on, when I started working as a school teacher, I became interested in what in science education we call "nature of science": what science is and how it is done. A colleague at my school recommended that I read a classic chapter by (my now good friend) Bill McComas. That was a key moment for me. What Bill did was to draw on philosophy of science to dispel various myths that I had also found to be popular among my students. That was a chapter with references to Popper, Kuhn, Lakatos, and this was when I began a more serious and systematic study of philosophy of science.

Then I got involved in the International History, Philosophy, and Science Teaching group, the journal *Science & Education*, and the rest is history.

Q- In your recent plenary talk at EvoKE-BCN 2021, you distinguished between *evolution* literacy and *evolutionary* literacy, and argued that the latter is of uttermost importance for science education. Can you briefly explain the difference between both concepts and make a sketch of your argument in favor of promoting *evolutionary* literacy?

R- Douglas Roberts has made a very useful distinction between two types of literacy relevant to science:

- *Science literacy:* Literacy relevant to processes and products of science, re-

lated to the content of science taught in classrooms (literacy about issues within science).

- *Scientific literacy:* Literacy relevant to questions that students may encounter as citizens and to the socio-ethical implications of scientific knowledge (literacy about the implications of science for society).

Based on this, we can distinguish between two types of literacy related to evolution:

- *Evolution literacy:* Literacy relevant to the evolution content taught in classrooms

- *Evolutionary literacy:* Literacy relevant to questions that students may encounter as citizens and to the socio-ethical implications of scientific knowledge

One might argue here that in any science domain one could distinguish between elements pertaining either to science or scientific literacy. Therefore, there does not seem to be any reason to make this distinction specific to evolutionary biology. But I respectfully disagree. Evolutionary theory can have direct implications for a person's worldviews, affecting beliefs such as racist ideas. Therefore, teachers need to be aware of these implications, and be confident and competent to discuss these in the classroom.

Q- Given that *evolutionary* literacy largely deals with non-strictly scientific issues, how do you think it can be best promoted in the classroom?

R- This is far from simple. Teachers would need to be aware of the respective issues: understand the science, its use and misuse, as well as the interpretations and the possible misinterpretations of the findings. There is a lot to digest here, but in my case this is what my books are for. With respect to evolutionary literacy, worldview and racism, my books Understanding Evolution (Cambridge University Press, 2020) and Ancestry Re-Imagined: Dismantling the Myth of Genetic Ethnicities (Oxford University Press, 2023) cover most, if not all, of the relevant topics.

Q- Besides those problems related to *evolutionary* literacy, what are the most pervasive problems for developing *evolution* literacy among students?

R- Students' preconceptions, which result from deeply-held intuitions, such as design teleology and psychological essentialism, which in turn make evolutionary theory seem counterintuitive. The key idea here is that intuitively we perceive the features of organisms in the same way that we ought to perceive the features of

artifacts: as designed for a purpose and as having fixed essences. As a result, the idea that evolutionary outcomes are contingent seems hard to accept.

This is why teachers at all levels need to be aware of students' preconceptions about evolution, which make it difficult to understand. These preconceptions need to be challenged explicitly during teaching, in order for students to undergo a change in their conceptions. Research has shown that this can be done by bringing them to conceptual conflict situations. These are situations to which students are brought where they can compare their preconceptions with those offered by the teacher, and subsequently come to realize that the latter are better in terms of their explanatory potential than the former.

Q- You have recently edited a book together with Tobias Uller (Lund University) entitled *Philosophy of Science for Biologists*. What do you think philosophy of science can contribute to the practicing biologist?

R- As we write in the Preface of that book, we do not know if Richard Feynman actually said that philosophy of science is as helpful to science, as ornithology is to birds. However, we have come across biologists, with prestigious grants and publications, who think that any philosophical discussion is a waste of time. The experienced researcher, they would say, know what has to be done; the inexperienced has to learn from the experienced ones in the lab or in the field. Whatever Kuhn or Popper said (they have not heard about Lakatos, or any philosopher after him), is irrelevant to the actual practice of science. Philosophy of science is, at best, a nice endeavor for retired scientists, if they decide to reflect upon their own career and work.

We believe that the topics covered in the book are some of the most useful ones for any biologist, especially the younger ones, to understand. These include what constitutes an explanation in biology; what biological knowledge is; and what the nature theories and models in biology is; how biology concepts are used and transformed; why it matters that many biology concepts are metaphors; how concepts contribute to scientific advancement; how conceptual analysis can contribute to scientific practice; the methods used in the life sciences; how biologists study the past and why this kind of work can be as solid as experimental science; what the basis of biological classification is; what the nature of scientific controversies in the biological sciences is; and what the relation is between facts and values in biological science.

This list is far from exhaustive, for sure, but we believe it is a good start.

Q- If you would have to say only one particular philosophical topic on which evolutionary biologists should be versed about given its importance to understanding how evolution works, what would you choose?

R- I do not think that there is any single philosophical topic that is more important than the others. But if the question were reframed and asked what would be the first topic to consider with young researchers and especially students, I am inclined to think that this should be the nature and structure of scientific explanation. Perhaps the most important, ultimate aim of science is to explain nature. But there are many different explananda in evolutionary biology, including patterns (such as those resulting from natural selection) and unique events (such as the Cretaceous-Paleogene extinction). The explanations required in each case are different, and so delving into their nature and structure is important.

Q- In one of your recent books, *Uncertainty: How it Makes Science Advance* (coauthored with Kevin McCain), you argue that uncertainty is an integral part of science, and that we should view it not as a weakness, but rather as a strength of science and scientific reasoning. However, you also argue that our educational system does not teach students how to effectively deal with uncertainty and accept is as a natural part of science. Instead, we tend to give students black or white answers to complex issues, and obviate the uncertainty that surrounds much of our scientific knowledge of the world. How do you think this mismatch between how science really works and how we teach it has affected public trust of science during COVID-19 pandemic? What communicative strategies should be pursued by scientists who wish to defend science against denialism yet do not misrepresent how science really works?

R- Psychologist Gerd Gigerenzer has cogently stated the problem: we teach our children the mathematics of certainty, Euclidean geometry, but not the mathematics of uncertainty, statistics. The problem is that in trying to establish students' trust in science, we present it as certain and authoritative when it comes to the natural world. It is true that science is the single best way to approach, study and understand the natural world. It is also true that the understanding produced is solid. However, there are always factors we cannot control, there are always details we do not or cannot know, and therefore there are always uncertainties. In fact, and climate science is a fine example here, the more we understand a phenomenon, the better we become aware of the related uncertainties.

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Therefore, I think that the simplest thing to do would be to help our children understand statistics and probabilities for a start. There are so many brilliant non-fiction books on these topics that could be used for teaching students. We also need to be explicit about the uncertainties inherent in science. A counter-argument here is that if we present these uncertainties to students, we might challenge their trust in science. Rather, the argument goes, we have to help them understand it first, and later deal with the uncertainties. But I disagree. I think it is a bigger challenge for students' trust in science to present it as (almost) certain, only for them to later realize that it is not. Rather, I believe that we have to explain both its strengths and its limitations right from the start.

Q- What projects are you currently working right now? Can you tell us a little bit about them?

R- I am currently developing a new research project for delving deeper in teachers' construal of the concepts of evolutionary biology. I am also working, as usual, on various books: editing a collection about myths related to Charles Darwin's life and work, with some stellar historians contributing; writing a book to challenge the stereotypical account of mendelian genetics taught in schools; and writing a book about science distrust.

Q- Do you have any advice to young evolutionary biologists who are also interested in the educational and philosophical implications of evolutionary theory?

R- Read, read, read!!! There are many books that are relevant and interesting, and at the same time accessible. Key thinkers who have also written accessible books are Michael Ruse and Philip Kitcher.